1 Dashboard Overview

The purpose of this project is to analyze a relational database of grocery sales to create an interactive business intelligence dashboard. It utilizes an end-to-end workflow, starting from data requirements gathering and data collection, followed by data cleaning and preprocessing to ensure consistency and accuracy across the dataset. Once the resulting dataset has been curated, exploratory data analysis was then performed to identify key patterns in sales performance, customer behaviour, and category distribution. This was complemented with efficient querying and aggregation of data to isolate the data in a way that allowed the creation of calculated metrics. Finally, the processed data was visualized through the use of pivot tables, KPI cards, slicers, and timelines, resulting in an interactive dashboard used for reporting sales trends, geographic performance, and category-level analysis.

2 Data Preparation and Aggregation

To support dashboard development and improve query organization, several SQL views were created to aggregate and structure the transactional sales data into analysis-ready formats. These views simplified the calculation of KPI metrics and visualizations by pre-grouping revenue, transaction counts, and category-level trends used throughout the dashboard.

2.1 Customer Sales Aggregation View

This SQL view was created to aggregate and structure transactional sales data at the customer level by joining records with product, category, customer, and geographic information. It generates a detailed dataset containing customer details, product details, pricing, discount application, and computed total sales value.

```
CREATE OR REPLACE VIEW sales_CUST_view AS
SELECT
    CONCAT(customers.firstname, ' ', customers.lastname) as
        customername,
    categories.categoryname,
    products.productname,
    products.price,
    sales.quantity,
    sales.discount,
    (products.price * sales.quantity) * (1 - sales.discount)
        as totalprice,
    sales.salesdate,
    customers.address as customeraddress,
    cities.cityname,
    countries.countryname,
    products.resistant,
```

```
    products.isallergic
FROM sales
LEFT JOIN
    products ON sales.productid = products.productid
LEFT JOIN
    categories ON products.categoryid = categories.
       categoryid
LEFT JOIN
    customers ON sales.customerid = customers.customerid
LEFT JOIN
    cities ON customers.cityid = cities.cityid
LEFT JOIN
    countries ON cities.countryid = countries.countryid
WHERE
    sales.salesdate IS NOT NULL AND
    sales.salesdate < DATE '2018-05-01';
```

2.2 Employee Sales Aggregation View

This SQL view was created to aggregate and structure transactional sales data at the customer level by joining records with product, category, employee, and geographic information. It generates a detailed dataset containing employee details, product details, pricing, discount application, and computed total sales value.

```
CREATE OR REPLACE VIEW sales_EMPL_view AS
SELECT
    CONCAT(employees.firstname, '␣', employees.lastname) as
       employeename,
    categories.categoryname,
    products.productname,
    products.price,
    sales.quantity,
    sales.discount,
    (products.price * sales.quantity) * (1 - sales.discount)
        as totalprice,
    sales.salesdate,
    cities.cityname,
    countries.countryname,
    products.resistant,
    products.isallergic,
    employees.birthdate as employeebirthdate,
    employees.gender as employeegender,
    employees.hiredate as employeehiredate
FROM sales
LEFT JOIN
    products ON sales.productid = products.productid
LEFT JOIN
```

```
    categories ON products.categoryid = categories.
        categoryid
LEFT JOIN
    employees ON sales.salespersonid = employees.employeeid
LEFT JOIN
    cities ON employees.cityid = cities.cityid
LEFT JOIN
    countries ON cities.countryid = countries.countryid
WHERE
    sales.salesdate IS NOT NULL AND
    sales.salesdate < DATE '2018-05-01';
```

2.3 Employee Sales Aggregation View

This SQL view was created to aggregate transactional sales data by month, category, and city in order to support trend-based analysis and dashboard visualizations. It generates summarized metrics such as total revenue, total quantity sold, sales count, and average order value, allowing monthly sales performance and category-level trends to be analyzed over time.

```
CREATE OR REPLACE VIEW sales_monthly_category AS
SELECT
    DATE_TRUNC('month', sales.salesdate) AS month,
    categories.categoryname,
    cities.cityname,
    SUM(products.price * sales.quantity * (1 - sales.
        discount)) AS total_revenue,
    SUM(sales.quantity) AS total_quantity,
    COUNT(*) AS sales_count,
    SUM(products.price * sales.quantity * (1 - sales.
        discount)) / NULLIF(COUNT(*), 0) AS avg_order_value
FROM sales
LEFT JOIN
    products ON sales.productid = products.productid
LEFT JOIN
    categories ON products.categoryid = categories.
        categoryid
LEFT JOIN
    customers ON sales.customerid = customers.customerid
LEFT JOIN
    cities ON customers.cityid = cities.cityid
WHERE
    sales.salesdate IS NOT NULL AND
    sales.salesdate < DATE '2018-05-01'
GROUP BY
    DATE_TRUNC('month', sales.salesdate),
    categories.categoryname,
    cities.cityname;
```

3 KPI Cards

This section provides a concise overview of overall business performance by summarizing key metrics, including customer activity, transaction volume, product movement, and revenue generation. These metrics are dynamically filtered based on the selected product category, enabling users to efficiently evaluate performance across different segments of the business.

3.1 Unique Customer Count

This metric calculates the total number of distinct customers who made purchases in the dataset.

```
SELECT
    categoryname,
    count(distinct(customername)) as count_unique_customers
FROM sales_cust_view
GROUP BY
    categoryname

UNION ALL

SELECT
    'All␣Categories',
    count(distinct(customername)) as count_unique_customers
FROM sales_cust_view
ORDER BY
    categoryname
```

Results Sample:

	categoryname character varying	count_unique_customers bigint
1	All Categories	93862
2	Beverages	93397
3	Cereals	93702

3.2 Total Quantity Sold

This metric calculates the total quantity of products that have been sold in the dataset.

```
SELECT
    categoryname,
    sum(quantity) as quantity_sold
FROM sales_CUST_view
GROUP BY
```

```
    categoryname

UNION ALL

SELECT
    'All␣Categories',
    sum(quantity) as quantity_sold
FROM sales_empl_view
ORDER BY
    categoryname
```

Results Sample:

	categoryname character varying	quantity_sold bigint
1	All Categories	80923708
2	Beverages	6808251
3	Cereals	8037396

3.3 Total Sales Made

This metric calculates the total number of sales that have been made in the dataset.

```
SELECT
    categoryname,
    count(*) as count_sales_made
FROM sales_empl_view
GROUP BY categoryname
UNION ALL
SELECT
    'All␣Categories',
    count(*) as count_sales_made
FROM sales_empl_view
ORDER BY categoryname
```

Results Sample:

	categoryname character varying	count_sales_made bigint
1	All Categories	6223697
2	Beverages	524104
3	Cereals	618237

3.4 Sales Revenue

This metric calculates the sales revenue in the dataset.

```
SELECT
    categoryname,
    sum(totalprice) as sum_total_sales
FROM sales_empl_view
GROUP BY
    categoryname

UNION ALL

SELECT
    'All␣Categories',
    sum(totalprice) as sum_total_sales
FROM sales_empl_view
ORDER BY
    categoryname
```

Results Sample:

	categoryname character varying	sum_total_sales numeric
1	All Categories	3989409391.26888
2	Beverages	337471895.43761
3	Cereals	393330435.15624

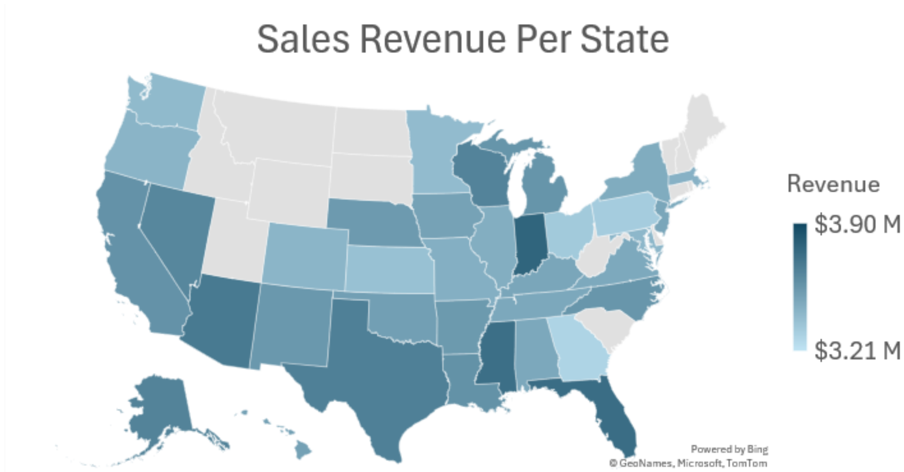
4 Charts

4.1 Sales Revenue Per State

This visualization displays total sales revenue across different states to highlight geographic sales distribution. A slicer that allows users to control which categories to include was incorporated to allow revenue comparisons across different product categories.

```
SELECT Month, SUM(SalesRevenue) AS TotalRevenue
FROM Sales
GROUP BY Month
ORDER BY Month;
```

Visualization:

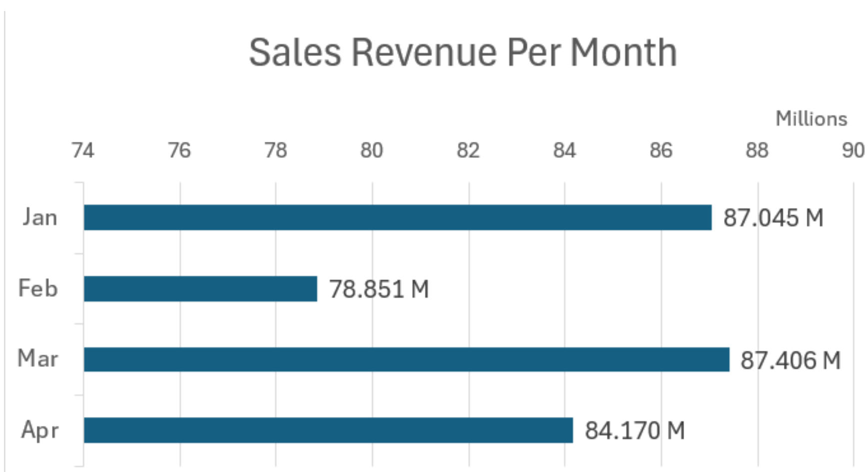


4.2 Sales Revenue Per Month

This visualization displays total sales revenue by month to analyze changes in overall sales performance over time. A slicer that allows users to control which categories to include was incorporated to allow revenue comparisons across different product categories.

```
SELECT Month, SUM(SalesRevenue) AS TotalRevenue
FROM Sales
GROUP BY Month
ORDER BY Month;
```

Visualization:

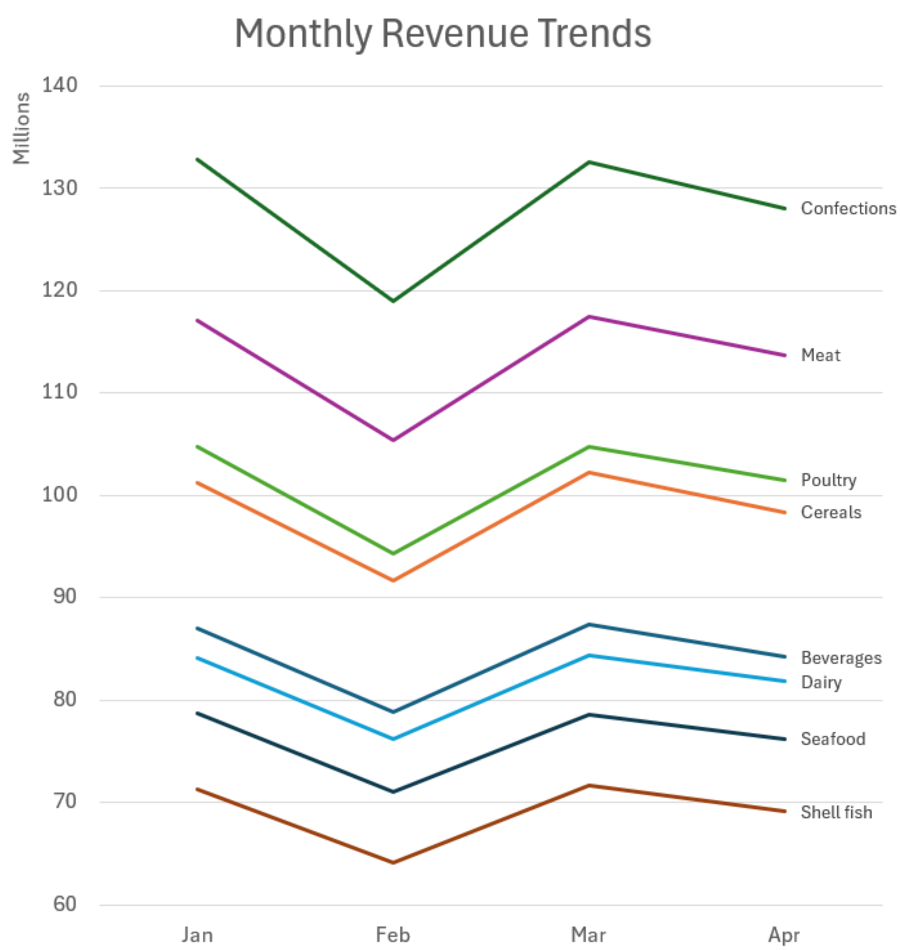


4.3 Monthly Revenue Trends

This visualization compares monthly revenue trends across product categories to identify changes in category-level performance over time. Both a category name slicer and a timeline slicer were incorporated to support filtering and trend analysis.

```
SELECT Month, SUM(SalesRevenue) AS TotalRevenue
FROM Sales
GROUP BY Month
ORDER BY Month;
```

Visualization:



5 Insights

The dashboard analysis highlights several consistent patterns across geographic, monthly, and category-level sales performance.

From the sales revenue per state visualization, overall revenue distribution remains relatively stable across different category selections. Most states maintain similar relative performance throughout the various conditions, with only minor shifts between categories. For example, California shows a slightly larger proportion of sales in categories such as seafood, dairy, and confections compared to others, though this variation is not strong enough to significantly alter its overall position relative to other states. Despite these small category-driven fluctuations, Mississippi consistently appears as the darkest region across all category filters, indicating that it remains the highest revenue-generating state regardless of product category. Overall, while minor variations exist between states under different filters, the general geographic distribution and ranking of regions remains largely consistent.

In the monthly sales revenue analysis, a more defined pattern emerges. January and March consistently show similar revenue levels across different categories, while April generally ranks as the second strongest month. February, however, shows a noticeable decline in revenue compared to the other months, indicating a clear dip in sales activity during that period.

The category revenue trend visualization shows highly similar trajectories across all product categories. While their magnitudes at each point vary based on the category, the shape and movement of the trends remain consistent, suggesting that category performance follows the same underlying seasonal pattern over the observed time period.

Overall, the results suggest that sales behaviour is more strongly influenced by time-based trends than by geographic or category-specific variation, with consistent monthly patterns driving most of the observed changes in revenue.